

Manish Tiwari

LLM Systems Engineer | AI Infrastructure | Open Source Contributor

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Projects

- SafeRun AI – Secure Sandbox Execution Platform** [Package] [GitHub] [Live] May 2026
- Built Docker-based sandbox executing untrusted AI-generated Python code in isolated containers with CPU, memory, and timeout limits.
 - Implemented AST scanner and YAML policy engine detecting dangerous imports, blocked calls, and infinite loops; classifies execution into **4 risk levels**.
 - Developed FastAPI backend with SQLite audit logging, optional LLM-powered explanations, and CLI + browser UI; distributed via **pip install saferun-ai**.
- Trimurti-LM: 4.2M Parameter Multilingual GPT** [Model] [Dataset] [Demo] [GitHub] Jan 2026
- Trained **4.2M** parameter GPT-style language model from scratch on **450K** multilingual (EN/HI/PA) sentences using custom SentencePiece tokenizer with byte fallback support.
 - Achieved perplexity **3.32** with training completed in **2.38 hours** on GTX 1650 4GB using FP16, gradient checkpointing, and gradient accumulation.
 - Published multilingual model, dataset, and Dockerized Streamlit frontend on HuggingFace.
- Production Model Drift Monitoring System** [GitHub] [Live] Feb 2026
- Built MLOps monitoring system detecting concept drift using statistical(KS) performance comparison with configurable alert thresholds and real-time scoring.
 - Built FastAPI async logging pipeline, Plotly dashboard, and Discord webhook alerting system.
 - Containerized deployment with Docker and automated CI/CD testing via Github Actions; achieved **<500ms** inference latency.

Experience

- Open Source Software Contributor** Global
- Hugging Face · CHAOSS/Augur · Moss +7 more* Jan 2024 – Present
- Contributed 32+ merged PRs across 10+ open-source organizations spanning AI/ML, developer tooling, testing, documentation, and bug fixes.
 - Collaborated with maintainers through 50+ code review discussions; delivered features, API enhancements, CI improvements, and automated test suites across Python repositories.
- Travintop.com** Remote
- Data Operations Intern* Dec 2024 – Feb 2025
- Built Python automation scripts to process and normalize 500+ travel package records for CMS upload workflows.
 - Analyzed influencer engagement metrics and structured operational datasets supporting marketing and listing operations.

Technical Skills

Languages: Python, SQL, C/C++, Bash
AI/ML: PyTorch, HuggingFace, Transformers, PEFT, LoRA, QLoRA, Tokenization, Fine-Tuning, LLM Training, Drift Detection
Backend/Infra: FastAPI, REST APIs, Async APIs, Docker, Linux, SQLite, GitHub Actions
Tools: Streamlit, Git&Github, Ollama, Plotly, SentencePiece, Scikit-learn

Education

IK Gujral Punjab Technical University Jalandhar, Punjab, India

Bachelor of Technology in Computer Science & Engineering; CGPA: 8.0 July 2023 – June 2027

Achievements

- Built and published Ollama libraries for Sarvam 30B and 105B multilingual LLMs with 100+ combined community pulls.
- Won Industrial Automation hackathon category for building “EnergiX Copilot,” an AI assistant for energy optimization and operational monitoring.
- Fine-tuned Mistral-7B and Qwen-14B using LoRA for dual-agent workflows at AMD Hackathon, IIT Delhi.